

NBD Focus Map

How to use Non-Brand Data without learning randomly.

What this is

Non-Brand Data is for people who want to improve in data and AI without getting lost in endless content.

This Focus Map turns the newsletter into a simple path. Choose one track, follow the cadence, ship one mini project, then switch.

How to use this

- Pick one track below and treat it like a short sprint.
- Stay on that track for 2–4 weeks. Do not switch halfway.
- Each week, follow the cadence and produce something you can show.
- At the end, ship the mini project. That is the point of the track.
- Once you ship, you can move to the next track.

Weekly cadence

- Plan for three sessions per week, around 60 minutes each.
- Every week, you should end with one visible artifact.
- It can be a note, a notebook, a small repo, or a short write-up.
- If you are not shipping anything, you are probably consuming too much.
- Keep it simple. Finish the track. Then switch.

Track A — SQL (for real analysis work)

This SQL Crash Course is a collaboration with Josep Ferrer (DataBites). Some lessons open on databites.tech.

What you will be able to do after this track

Pull the right data, define metrics clearly, combine tables safely, write modular queries, and improve performance when queries slow down.

Read in this order (5 posts)

1. Basic Commands (SELECT, FROM, WHERE)
<https://www.nb-data.com/p/4-sql-basic-commands>
Outcome: You can correctly filter and shape a dataset.
2. Aggregate Functions (SUM, AVG, COUNT, etc.)
<https://www.nb-data.com/p/6-aggregate-functions>
Outcome: You can compute KPIs and grouped metrics that stay consistent.
3. JOINS (LEFT, RIGHT, INNER, FULL)
<https://www.databites.tech/p/7-joins-left-right-inner-and-full>
Outcome: You can combine tables without inflating counts or duplicating rows.
4. Common Table Expressions (CTEs)
<https://www.databites.tech/p/12-common-table-expressions-ctes>
Outcome: You can write modular SQL and build queries in clear steps.
5. Query Optimization
<https://www.nb-data.com/p/20-query-optimization>
Outcome: You can reduce runtime and make queries more production-friendly.

Build this mini project (one weekend)

Pick one business question and answer it end-to-end using the five concepts above.

Choose one:

- Funnel drop-off: “Where do users abandon onboarding?”
- Cohort retention: “How does 30-day retention look by signup week?”
- Revenue concentration: “Which products drive repeat purchase?”

Deliverables:

- One SQL file with sections: `prep` → `metrics` → `final_output`
- A short report with 3–5 insights
- One table per insight
- A “Next steps” section with 3 bullets

What “good” looks like

- The query reads like steps, not one long block
- Each metric has a one-sentence definition
- Joins do not change row counts unexpectedly
- The output is decision-ready, not just correct

Track B — Python + ML (for practical modeling)

What you will be able to do after this track

Prepare data cleanly, build a baseline model, choose sensible algorithms, tune with intent, and explain outcomes in business language.

Read in this order (5 posts)

1. Use Pandas Like a Pro
<https://www.nb-data.com/p/use-pandas-like-a-pro>
Outcome: You can prepare data faster and avoid common pandas mistakes.
2. 6 Best Practices for Machine Learning
<https://www.nb-data.com/p/6-best-practices-for-machine-learning>
Outcome: You can frame the problem well and avoid building the wrong model.
3. Common Classification Machine Learning Algorithms
<https://www.nb-data.com/p/nbd-lite-2-common-classification>
Outcome: You understand the main classifier families and when to use them.
4. Why Your Hyperparameter Tuning Is Wasting Time?
<https://www.nb-data.com/p/why-your-hyperparameter-tuning-is>
Outcome: You tune with intent and save compute without losing quality.
5. Build Machine Learning Explainability Explainer with SHAP and LLM
<https://www.nb-data.com/p/build-machine-learning-explainability>
Outcome: You can justify predictions in a way that stakeholders understand.

Build this mini project (one weekend)

Build one ML workflow from raw data to evaluation.

Deliverables:

- One notebook: EDA → features → model → evaluation
- One baseline model and one improved model
- One metric choice with a short justification
- A “Next steps” section with 3 bullets

What “good” looks like

- The notebook is reproducible
- Metric choice matches the use case

- Baseline comes first
 - The explanation is usable by non-technical readers
-

Track C — RAG / AI Systems (for modern AI engineering)

What you will be able to do after this track

Build a baseline RAG system, improve retrieval quality, add safeguards, then evaluate and monitor it with a production mindset.

Read in this order (5 posts)

1. Simple RAG Implementation With Contextual Semantic Search
<https://www.nb-data.com/p/simple-rag-implementation-with-contextual>
Outcome: You can build a working baseline RAG pipeline.
2. 9 Chunking Strategies to Improve RAG Performance
<https://www.nb-data.com/p/9-chunking-strategies-to-improve-rag>
Outcome: You can split documents in ways that improve retrieval.
3. RAG Reranking To Elevate Retrieval Results
<https://www.nb-data.com/p/rag-reranking-to-elevate-retrieval>
Outcome: You can improve relevance beyond vector similarity alone.
4. Building Guardrail Around Your RAG Pipeline
<https://www.nb-data.com/p/building-guardrail-around-your-rag>
Outcome: You reduce common failure modes and unsafe outputs.
5. RAG Evaluation Monitoring and Logging with Opik
<https://www.nb-data.com/p/rag-evaluation-monitoring-and-logging>
Outcome: You can evaluate retrieval and generation, then track quality over time.

Build this mini project (one weekend)

Build a small RAG demo that answers questions from a document set.

Deliverables:

- 5–20 documents as your knowledge base
- A retrieval step (vector or hybrid)
- A prompt template that cites sources
- A test set of 10 questions
- An evaluation log: grounded or not, plus why

What “good” looks like

- Answers cite sources
 - Retrieval failures are visible
 - You keep a test set and a log
 - You can explain the trade-offs
-

Which track should you pick?

Pick based on your current bottleneck:

- If you struggle to extract insights from data, pick SQL
- If you struggle to build end-to-end modeling work, pick Python + ML
- If you want to build modern AI systems, pick RAG / AI Systems

Suggested next step

After you finish one track, you will have a reusable artifact: a notebook, a repo, a write-up, or a portfolio entry.

After you finish a track

You should now have a reusable artifact: a notebook, a repo, a write-up, or a portfolio entry.

Next steps:

1. **Sharpen it** with a portfolio rubric, so you know exactly what to fix and in what order.
2. **Repeat the process** with the next track, using the same cadence and shipping rule.